

Traceable In-Air Signature 3D Restoration Record Structure and In-Air Dominant Hand Biometric Based on Dynamic Time Warping Algorithm

Yuheng Guo

University of Tokyo

yuhengguo@satolab.itc.u-tokyo.ac.jp

Hiroyuki Sato*

University of Tokyo

schuko@satolab.itc.u-tokyo.ac.jp

Abstract—The portable device in-air signature is developing at a fast speed for the past decades because of the advantage of being remote and traceless, but it has gained limited application in the business field. Lacking supervision during the enrollment phase is one of the security risks for in-air signature recognition. We propose a traceable in-air signature structure based on 3D signature restoration for dominant and non-dominant hand in-air signatures. We designed a dominant hand signature recognition model based on the dynamic time warping (DTW) algorithm to reduce the security threat during the enrollment phase of the in-air signature. By applying the DTW model to, in total, 80 in-air signature dominant and non-dominant hand samples collected through the self-designed application, the model can achieve 100% accuracy if more than 6 training samples are taken for each subject and 0% EER for the available participants.

Index Terms—In-air signature recognition, sensor, security, privacy, dominant hand recognition

I. INTRODUCTION

With the development of modern mobile phone sensors, the accuracy of accelerometers and gyroscopes has been improved to a brand new level, which enables a new field of application design related to mobile phone sensors. Mobile phone in-air signature has been a new substitution for traditional handwritten signatures, given the advantage of being remote, trace-less, and convenient [1]. Especially in the era of the global pandemic, social distancing is regulated by various global organizations and governments [13]. Being able to sign signatures in the air using people's portable devices like the mobile phone is crucial for the normal conduction of businesses [14]. Nevertheless, within the previously proposed in-air signature procedure in many types of research, security issues exist during the enrollment phase of the in-air signature. Unlike traditional signatures signing, people have direct eye supervision throughout the whole process and can easily distinguish the signer's dominant hand. The in-air signer could potentially use their non-dominant hand to finish the in-air signature enrollment process while using their dominant hand to sign daily business contracts. Lacking the ability to detect and trace back the dominant and non-dominant hand in-air signature records of individuals could pose regulation breaches and legal ambiguities since the recognition structure of the dominant hand is not defined by most of the applications. In this paper, we propose a novel tractable dominant-hand

recognition structure by restoring the 3D in-air signature records and designing a dominant hand biometric system based on DTW to improve the security of the in-air signature process. The model can achieve 100% accuracy if more than 6 training samples are taken for each subject and 0% EER for the available participants.

II. RELATED WORK

In-air signature biometrics has been researched extensively in the past decades and various method has been tested to improve the accuracy of the recognition model. As in Abdulhalik Oğuz's research [5], the research used 387 mobile phone devices of various types to collect individual information, including the accelerometer data, the voice, and other information that can be collected from the phone. 3000 samples have been collected for the analysis. The model used in the research is k nearest neighbor (KNN) and the randomized neural network (RNN) which gives 99.994% for KNN and 99.97% for RNN in terms of identifying individuals. Gonzalo Bailador et al. [1] used the Hidden Markov Model, Bayes classifiers, and dynamic time warping algorithms for the analysis of mobile phone in-air signatures. 96 participants are invited to the experiment, from whom 753 valid signatures are collected. The signatures collected are divided into two categories, which are genuine and imposter signatures. As a result, the best performance EER for the HMM model is $5.96\% \pm 0.55$, for the Bayes model is $14.09\% \pm 0.37$, and $4.58\% \pm 0.51$ for the DTW model. In Elyoenai Guerra-Segura et al.'s research [6], 2000 samples are collected from 100 subjects with 10 genuine and 10 imposter signatures. The imposters are further divided into zero-effort and active imposters. The model used in the experiment is the Least Square Support Vector Machine, which gives an equal error rate of 0.25% by zero-effort attackers and 1.20% by active attackers. Souvik Ghosh et al. [3] used deep learning network models to improve the accuracy of recognition. The model used in the research is called ST-SNN. ST-SNN is mainly a Siamese Neural Network which is consisted of a Convolutional Neural Network (CNN) and Long Short-Term Memory networks (LSTM). The dataset used in research is an open-source in-air signature dataset collected by S.K. Behera et al. [4], which consists of 80 participants and 1600 in-air

signatures. As a result, the true positive rate is 94.63% and the False Acceptance Rate is 4.1% for the model. In Javier Guerra-Casanova's research [7], Linear discriminant analysis (LDA) can be used for sequential signals recognition and it has been applied to in-air signatures in the experiment. 120 gestures have been collected but the EER result is higher than expected. After dropping out some of the gestures, 4.01% of EER can be achieved by using the LDA model. S. Wang et al. [9] analyze the phone's orientation effect on the sensor's accuracy during the process of in-air signature recognition. The research uses K-means and Random Forest for the evaluation of continuous authentication on mobile phones. The mobile phone authentication is all based on a support vector machine. The author also proposed adaptive models to reduce the effect of orientation on mobile sensors. By using the adaptive model, the authentication gives 15.69% FAR and 2.25% FRR. Given the convenience of the in-air signature being remote and long distance, it would be better to be able to sign in the air without using any external devices. Yuxun Fang et al. [2] used camera-captured videos as a reference to recording individuals' in-air signatures from videos. They used the DTW and Fast Fourier Transform (FFT) for the preprocessing of the data and used Gaussian distributions to categorize signatures into various sets. They collected the signature information from 14 subjects and in a total of 560 signatures for the analysis. The model shows 2.86% FRR and a 1.90% FAR for the in-air signature recognition. Yuxun Fang's in-air signature 2D restoration is considered closest to our approach but we used the original data collected from sensors and no formal structure for traceable in-air signature has been proposed in Yuxun Fang et al.'s experiment.

III. SYSTEM MODEL

The Fig. 1¹ demonstrates the proposed traceable in-air signature 3D restoration record structure and the dominant hand recognition model. The participants are required to input their in-air signatures by signing their name in the air using the mobile phone software developed through Android SDK of 20 rounds of signature, 10 rounds of dominant hand signature, and another 10 rounds of non-dominant hand signature, are taken into analysis. Those 20 rounds of signature signals are divided and sent to the 3D signature restoration model in which a collection of dominant and non-dominant hand signatures are kept as legal records. Meanwhile, 20 signature signals will be fed into the dynamic time warping model for recognition. After training, the model can distinguish the new signature from the end users which can tell if the new in-air signature from the user is signed by the dominant or the non-dominant hand. Only the dominant hand in-air signatures will be regarded as a valid legal representation of the juridical person. Every time a new signature is used for contract purposes, the new signatures

¹Some of the symbols in graphs is from flaticon. Graph created based on drawio.

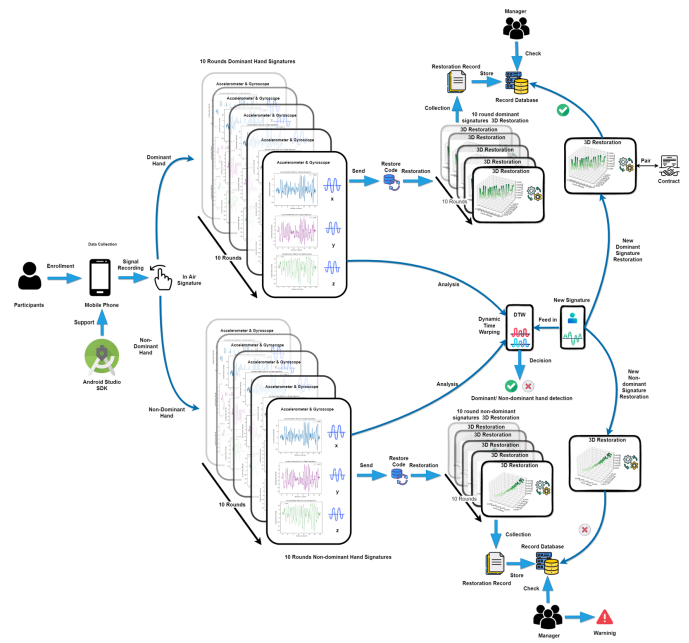


Fig. 1. Dominant hand recognition system structure

will also be stored in the database paired with the contract. When legal conflicts happen for the contract, the plaintiff can apply for verification from the database manager who has the access to the database and can view both the dominant and non-dominant hand in-air signatures from the signer and export 3D signature restoration together with the denial or approval data history of the DTW model.

IV. PROPOSED METHOD

A. Data Acquisition

In this experiment, we collected the in-air signature using mobile from 4 participants. Each participant is requested to sign their 3D signature in the air using the mobile phone device just like what they would do for handwritten signatures. For participants, they sign their signature 10 rounds by using their dominant hand and another 10 rounds using their non-dominant hand. By clicking on the start signature button when they are ready and clicking on the finish signature button when they finish. The temporary 3D signature data will be stored inside the application. Then the application will direct participants to the final page where they input their name information, the number of rounds for the signature, and toggle the dominant or non-dominant button on the interface. Once they finish inputting the information, the temporarily stored data will be sent automatically to the designated email for analysis. All personal information gathered in the experiment has gained consent from the participants and all the data are for the research purpose only.

TABLE I summarizes the configuration of the mobile phone used in the experiment, to replicate the experiment, it would

TABLE I
Configurations of the mobile phone used in the experiment.

Configuration List	
Configurations	Information
SoC	Qualcomm Snapdragon 480
CPU	2*2.0 GHz Cortex-A76,6*1.8 GHz Cortex-A55,Cores:8
GPU	Qualcomm Adreno 619,950 MHz
RAM	4 GB,1800*2400 pixels,24 bits

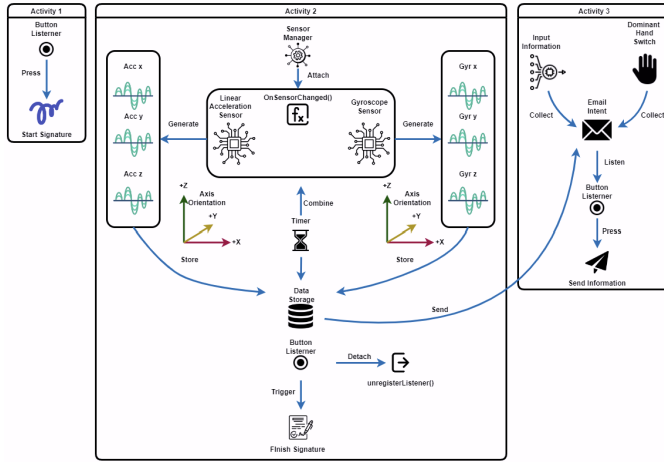


Fig. 2. In-air signature collection application structure based on Android Studio SDK

be ideal to use the configuration indicated in this table. The in-air signature data are collected based on the application we developed through the Android Studio SDK on mobile phone. The whole application structure can be divided into three main activities and four interfaces as shown in Fig. 2. The first activity demonstrates the guidance page and shows the relative information for the in-air signature collection, which is consisted of a button listen controller. Once the button is pressed the signature collection process begins. The application will then be directed to the second activity. The sensor manager works as the kernel for the linear acceleration sensor and the gyroscope sensor. Once the sensor manager has been activated, both the accelerometer and gyroscope will be attached to it throughout activity two. The x,y, and z coordinate information will be recorded together with the synchronized time stamps in the storage. As shown in the TABLE II,

TABLE II
Eight features were collected from the data collection application.

Feature Collection	
Symbol	Feature
Acc x	X-axis linear acceleration
Acc y	Y-axis acceleration
Acc z	Z-axis acceleration
Acc t	Acceleration time stamp
Gyr x	X-axis angular velocity
Gyr y	Y-axis angular velocity
Gyr z	Z-axis angular velocity
Gyr t	Angular velocity time stamp

there are in total 8 features collected for each signature. The x,y, and z coordinate positive directions defined by Android Studio are also illustrated in the Fig. 2. More specifically, the Android Studio axis coordinate system defines the positive x-axis direction as the horizontal line from the left to the right when a user is holding a phone. The positive y-axis direction is defined as the direction that goes into the paper. The positive z-axis is defined to point toward the user's head. No matter how the orientation of the user changes, the coordinate will not change as long as the user is holding the phone. The unit for the accelerometer and gyroscope are all meters per second squared while the unit for the timer is in milliseconds. The data information is stored in the format shown in the TABLE III.

TABLE III
Sample data from the collection application. The precision is 7 digits after the decimal, omitted here in the table.

Sample Data Format				
sensor	x-axis	y-axis	z-axis	Time
Acc	-5.48	-5.95	1.45	1086
Gyr	3.547	-4.92	-3.80	1090

TABLE IV demonstrates the sensor sampling frequency that can be adjusted by tuning the *SENSOR_DELAY* parameter. The table shows the available sampling rate for the application. For this experiment, we choose *SENSOR_DELAY_NORMAL* for both the accelerometer and gyroscope which is 215-230ms per sample. The sensors will detect the sampling request information by calling the function *OnSensorChanged()*. Once the participant has finished the signature, the finish signature button will be pressed. Both the accelerometer and gyroscope will be detached from the sensor manager to save the phone energy and improve the software performance. Then the participants will be directed to the third activity in which their information will be collected, including their name, the number of rounds, and the category of hand. The third activity is designed with the email intent object which could access the temporarily stored data information and send it to the designated email address automatically. In Fig. 3 ranging from the left to right represents the 4 interfaces in the collection application and the last interface shows the data gathered for one signature.

TABLE IV
Available sampling parameters and sampling rate for Android Studio SDK

Available Sampling Rate		
Sensor	Parameter	Sampling Rate
Acc	FASTEST	18-20 ms
Acc	GAME	37-39 ms
Acc	UI	85-87 ms
Acc	NORMAL	215-230 ms
Orientation	FASTEST	16-17 ms
Orientation	GAME	37-39 ms
Orientation	UI	77 ms
Orientation	NORMAL	224-225 ms

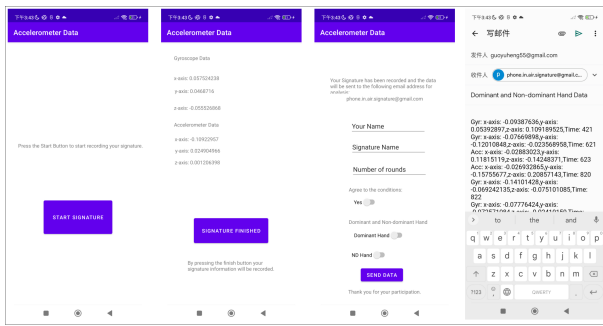


Fig. 3. In-air signature collection application interface

B. Data Analysis

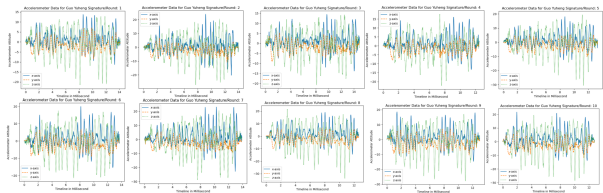


Fig. 4. Ten rounds of dominant hand acceleration signals. The y-axis of the graph shows the altitude of the sensor in meters per seconds squared. The blue line represents the x-axis acceleration, the yellow dashed line shows the y-axis acceleration, and the green dashed line represents the z-axis.

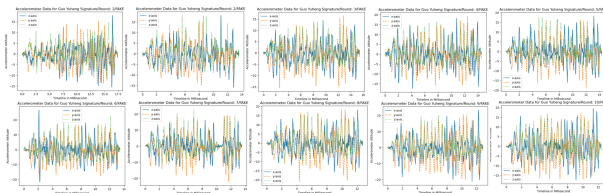


Fig. 5. Ten rounds of non-dominant hand acceleration signals. The y-axis of the graph shows the altitude of the sensor in meters per seconds squared. The blue line represents the x-axis acceleration, the yellow dashed line shows the y-axis acceleration, and the green dashed line represents the z-axis.

Once the data information has been gathered, we are interested in finding the pattern of the data and which feature contains most of the information that can help us distinguish dominant and non-dominant hand in-air signatures. More specifically, the x, y, and z-axis signals are extracted and compared through multiple rounds of signatures and the 3D representation of those signals is also illustrated visually to give a clearer impression. As shown in the Fig. 4, 10 rounds of the dominant hand acceleration signals are drawn for comparison. The x-axis of the graph shows the time converted from millisecond to second. From the graph, we could observe that even though 10 rounds of dominant hand in-air signatures have subtle differences, they overall show similar patterns. The position of the peaks and valleys reside in similar timeline locations. The x-axis and z-axis show much larger fluctuation while the y-axis is relatively stable. The reason is that when people sign in-air signatures, they tend to imagine drawing

on 2D paper. Therefore, the x-axis and z-axis may give rich information in determining the dominant hand. Similarly, in the Fig. 5, for 10 rounds of the non-dominant hand acceleration signal graph, the signal waves show much larger fluctuation compared with the dominant ones, especially for the y-axis.

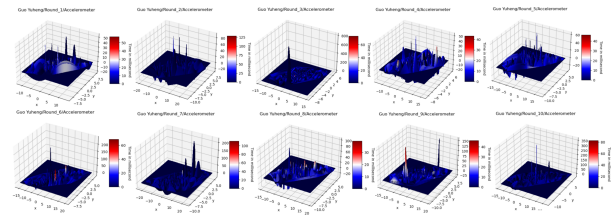


Fig. 6. Ten rounds of 3D dominant hand acceleration signals

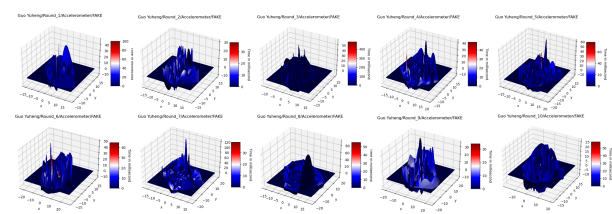


Fig. 7. Ten rounds of 3D non-dominant hand acceleration signals

To better illustrate and observe the difference between the dominant and non-dominant in-air signature, 10 rounds of each signature are drawn in three-dimensional graphs as shown in the Fig. 6 and Fig. 7. The time variable is represented from the deep blue color to the deep red color. The 3D representation of the signal shows a larger variance but the shape of the curve for the dominant hand (right hand in this sample) in the Fig. 6 shows that they all range from the first quadrant to the third quadrant, which complies with the subject's right-hand writing habit. Similarly, in the Fig. 7, despite the shape variance, the tendency all goes from the second quadrant to the fourth quadrant. The dominant hand and non-dominant hand could exhibit huge tendency variance, which confirmed our expectation that if a person uses the non-dominate hand for signature enrollment while using the dominant hand for signing a contract, the system won't be able to categorize them into the same Juridical Person.

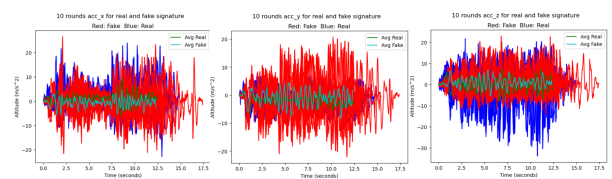


Fig. 8. Ten rounds accelerometer signals dominant and non-dominant hand comparison. The blue lines represent 10 rounds of dominant signature signals, and the red lines represent 10 rounds of non-dominant signature signals. The cyan line represents the average dominant signal. The green line represents the average non-dominant signal.

In Fig. 8, the accelerometer comparison and Fig. 9, the gyroscope comparison, the amplitude of average signals for

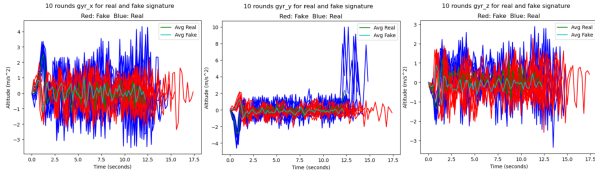


Fig. 9. Ten rounds gyroscope signals dominant and non-dominant hand comparison. The blue lines represent 10 rounds of dominant signature signals, and the red lines represent 10 rounds of non-dominant signature signals. The cyan line represents the average dominant signal. The green line represents the average non-dominant signal.

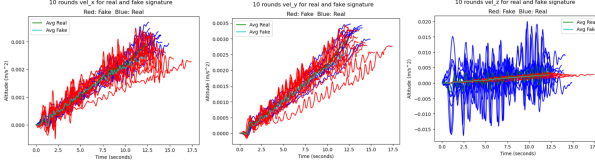


Fig. 10. Ten rounds velocity signals dominant and non-dominant hand comparison. The blue lines represent 10 rounds of dominant signature signals, and the red lines represent 10 rounds of non-dominant signature signals. The cyan line represents the average dominant signal. The green line represents the average non-dominant signal.

dominant and non-dominant in-air hand signatures not only exhibit a mismatch in the amplitude but also have reversed peak and valley. Nevertheless, for the Fig. 10 of the linear velocity signal comparison, the z-axis signals show tremendous differences and we need to consider those features in dynamic time warping calculation.

Algorithm 1 Dynamic Time Warping Algorithm [8]

1. Input: series $u = \{u_1, u_2, \dots, u_{T_u}\}$ series $v = \{v_1, v_2, \dots, v_{T_v}\}$

2. Base conditions:

$$\begin{aligned} g(0, 0) &= 0 \\ g(1, 1) &= d(u_1, v_1) \cdot w_D \\ g(i, 0) &= \infty \quad \text{for } 1 \leq i \leq T_u \\ g(0, j) &= \infty \quad \text{for } 1 \leq j \leq T_v \end{aligned}$$

3. Recursive relation:

$$g(i, j) = \min \begin{cases} g(i, j-1) + d(u_i, v_j) \cdot w_V \\ g(i-1, j-1) + d(u_i, v_j) \cdot w_D \\ g(i-1, j) + d(u_i, v_j) \cdot w_H \end{cases}$$

for $1 \leq i \leq T_u$ and $1 \leq j \leq T_v$

4. Output: DTW-distance $D(u, v) = k(w) \cdot g(T_u, T_v)$

Algorithm 1 explains the programming algorithm of dynamic time warping, in which two sequential signals are taken as the input for the model [8]. In our case, two sequential in-air signature signals are taken into consideration at each

time of the calculation. For each participant, we have 20 in-air signatures, which gives us in a total of 400 calculations of the dynamic time warping distance. We build a dynamic time warping cost matrix whose dimension complies with the dimension of two signals and four based conditions are needed to initialize the cost matrix. The zero origins are initialized to zero and the second element of the second row and of the second column is initialized as the starting distance of each signal. For the first row and first column, each element is initialized to be an infinitely large number. Then, moving from the origin, each cell of the DTW cost matrix is set to the minimum value of the neighboring 4 element cells with tuning coefficients. Then the final distance is the last element cell of the whole cost matrix. Going backward from the last cell to the initial cell, a warping path is calculated by connecting the minimum element cell which generates the shortest path to the origin of the cost matrix. The warping path tells us the alignment among different points of two sequential signals. Unlike the transitional point-wise alignment, multiple elements being equal in the horizontal and vertical direction of the cost matrix indicates one-to-many alignment among two signals.

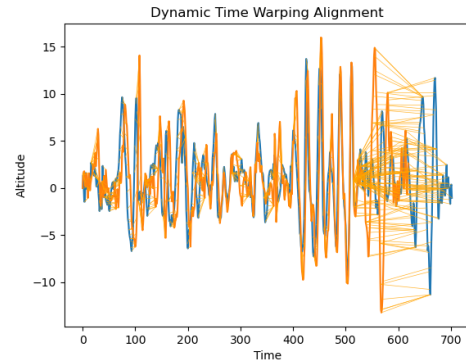


Fig. 11. Signature alignment for signature signals belonging to the same category using dynamic time warping. The orange and blue lines illustrate two signature signals belonging to the same category but of different time lengths.

Fig. 11 shows the alignment of the two linear acceleration signals. DTW algorithm can align the similar peaks and valleys of two signals as shown by those yellow lines. The Fig. 12 demonstrates the visual representation of the dynamic time warping cost matrix in which the two signature signals are drawn at the left and the top part of the graph and the sample signals belong to the same signature category. The top left corner of the graph shows the DTW distance value between those two sample signals, which is 34.6936. As explained in the algorithm part of dynamic time warping, the element cost is depicted as ranging from blue to light yellow, with the blue being lower cost and the yellow being higher cost. The red line is the warping path of the DTW cost matrix, two perfectly matching signals give a straight warping path from the origin to the last cell of the cost matrix. The blue parts of the graph are all surrounded around the warping path, which indicates a higher cost of alignment while the yellow part is all far away

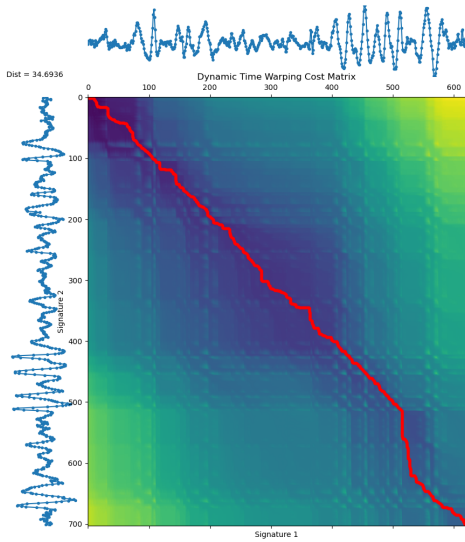


Fig. 12. Dynamic Time Warping Cost Matrix

from the warping path indicating a higher cost.

C. Three Dimensional Data Restoration

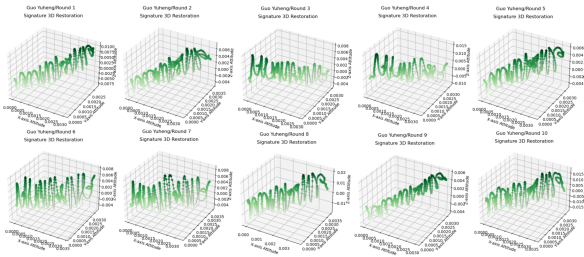


Fig. 13. Ten rounds of 3D dominant hand in-air signature Restoration

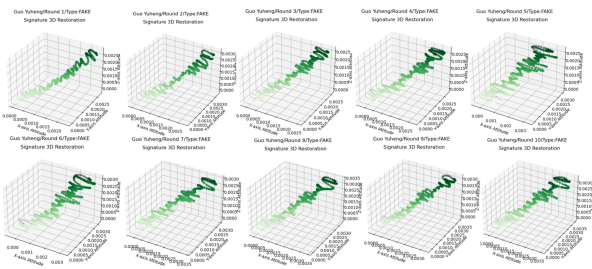


Fig. 14. Ten rounds of 3D dominant hand in-air signature Restoration

Given the in-air signature has the advantage of being traceless, the signer could sign and send the in-air signature digitally. Nevertheless, being traceless also means that there is no way of tracing whether the in-air signature is signed by a dominant or non-dominant hand in an explicit way, which can be a security threat to the business and legal contract. Therefore, being able to reconstruct the 3D restoration of the in-air signature is crucial and it can be helpful in terms of working as an important record for identification. Aligning

the in-air signature data with the visible 3D signature restoration can reduce the legal ambiguities in the business when two parties use the in-air signatures to sign their contract. As explained in the system model section, in our proposed structure, the restoration of 3D in-air signatures could be kept in a centralized database to trace back the signature history and work as a legal backup when conflict happens. Fig. 13 and Fig. 14 demonstrate the recovered dominant and non-dominant 3D in air signatures respectively. During the restoration process, multiple engineering obstacles have been solved. Firstly, the mobile phone calculates acceleration based on all the force applied to the phone, including gravity. The gravity-included data exhibit abnormally high negative acceleration for the z-axis during the signing process. Not only for the z-axis, the gravity will also be applied to other axes during rotation. During the restoration of the signature, the effect of gravity is needed to be eliminated. Otherwise, the gravity will affect the whole data set and can't be normalized, which will result in that the restored signature will turn to a straight line. In this experiment, we used the Android Studio *TYPE_LINEAR_ACCELERATION* parameter that can eliminate the gravity's effect by combining the angular velocity together with the linear acceleration but sacrificing a certain degree of accuracy. Secondly, the unit of the accelerometer in the Android Studio is meters per seconds squared and the timer in the JAVA library is in the millisecond, if the unit is not consistent, the restoration will be out of scale. Thirdly, the accelerometer and the gyroscope x,y,z, and t lists are needed to have an initial element zero to start with, otherwise, the result is not visible. In the end, the accelerometer will introduce inevitable noise to the system. After taking two integrals of acceleration with respect to time, we are expected to get the location signal that can be used for restoration, but the real meaningful location signals are buried within the noise and the 3D position signal will become a single smooth line. After testing, the best solution is to use the velocity vector to restore the 3D signatures. Normalization is also needed to be applied to three axes respectively and there are two main normalization methods, one is with respect to the summation of the list and the other one is with respect to the maximum element in the list. After testing, the summation normalization can better restore the 3D signature, which is also used as the normalization method for the DTW model covered in the result section.

V. RESULT

In our research, we constructed a 20 by 20 DTW distance matrix for each participant in which the first 10 rows and columns all belong to the dominant hand signatures while the last 10 rows and columns all belong to the non-dominant hand signatures. The distances among each signal are calculated in the DTW distance matrix, which in total gives us 400 distance elements for each participant. During the experiment, 8 features are collected for each signature as shown in the TABLE II, which means we need to calculate 8, 20 by 20, DTW distance matrix for each signature. As shown in the Fig. 15, by

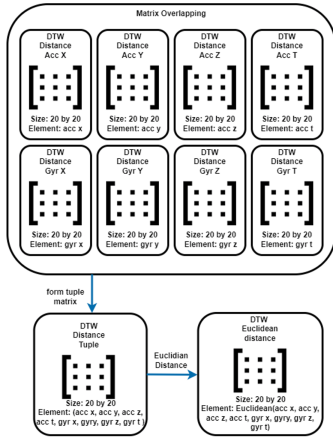


Fig. 15. DTW euclidean distance matrix calculation

overlapping those 8 matrices into one 20 by 20 tuple matrix with the elements being tuples containing 8 feature distances. The final DTW distance matrix is calculated by applying the euclidean distance to each of the tuple elements.

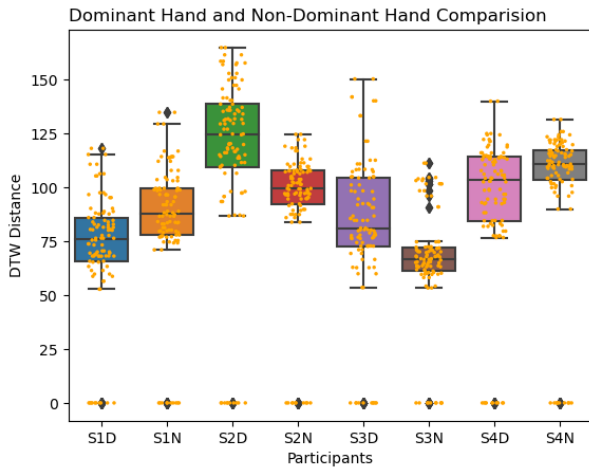


Fig. 16. The dominant hand and non-dominant hand comparison box plot for 4 participants. SX(D/N) means the number X participant's dominant(D)/non-dominant signature(N).

As shown in the Fig. 16, the dominant hand and non-dominant hand DTW distance variations among 4 participants is depicted as box plot. Those two categories show large differences in terms of the distribution of DTW distances, which explains the reason why DTW can distinguish one set of signatures from the other with high accuracy. Notably, the graph does show a non-intuitive phenomenon. During the planning phase of the experiment, the large variation difference between the dominant and non-dominant is expected but the non-dominant hand is expected to show a larger variation. Nevertheless, the data exhibit that the dominant hand signatures show much larger variation compared with the non-dominant hand signatures. The reason why dominant hand signatures show much larger variance compared

with the non-dominant hand signatures can be explained by C.A.ARMSTRONG et al. [12]. The research performed a strength comparison between the dominant and non-dominant hand using Maximum voluntary contraction (MVC) and 3% of strength differences are found for the right-handed subjects, which means that individual's dominant hand is much more flexible and can generate more signature variations compared with the relatively rigid non-dominant hands.

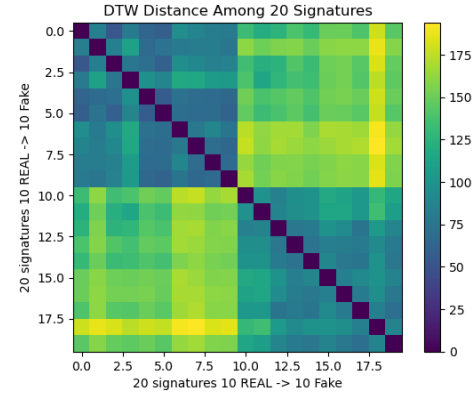


Fig. 17. DTW distance among 20 signatures with the first 10 signatures being the dominant hand and the last 20 signatures being the non-dominant hand

Fig. 17 is one of the DTW distance matrices for participants, each pixel represents a distance calculated by DTW, with the deep blue color representing a smaller distance value while yellow represents a larger distance value. The first and the third quadrants are the quadrants where the distance of dominant and non-dominant signature distances are. The second quadrant is the distances among 10 dominant hand in-air signatures while the fourth quadrant is the distances of 10 non-dominant signatures. The first and the fourth quadrant shows a clear color difference compared with the rest of the two quadrants, which indicates that DTW distance can be an efficient tool for distinguishing the dominant and non-dominant hand in-air signatures. In order to categorize 20 signatures, we need to find a reference signature signal, which is determined by calculating the overall distance of each signature among the rest of the signatures within the same category. Therefore, for each participant, two reference signatures are determined through 6 training samples. The rest of the signatures, dominant and non-dominant signatures, are testing samples which are categorized by calculating the relative DTW distance with the reference signature. To tune for the best performance, coefficient M_{dominant} and $M_{\text{non-dominant}}$ are added to better calculate the decision boundary.

False acceptance rate(FAR) and false rejection rate(FRR) are widely used in many biometric types of research, and we use them to empirically examine the performance of the DTW algorithm on our data set. As shown in (1) [11], FAR concerns the number of imposter samples that are categorized into the real samples while the FRR represents the rate when the real

samples are categorized into the imposter samples.

$$\begin{aligned} FAR &= \frac{|\{s \mid q \in \text{Sig}_{\text{Dominant Hand}}, s \geq \Delta\}|}{|\text{Sig}_{\text{Dominant Hand}}|} \\ FRR &= \frac{|\{s \mid q \in \text{Sig}_{\text{Non-dominant Hand}}, s < \Delta\}|}{|\text{Sig}_{\text{Non-dominant Hand}}|} \end{aligned} \quad (1)$$

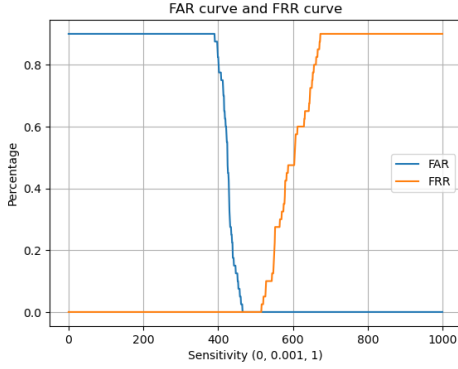


Fig. 18. False acceptance curve and the false rejection curve

We take 6, 3 dominant and 3 non-dominant, in air signature samples for training with regard for each participants and 14 samples for testing. As shown in the Fig. 18, the false acceptance rate curve stays in the high position starting from the 0 sensitivity and drops down steeply during the 0.4 to 0.5 sensitivity range. Similarly, for the false rejection rate curve, it stays in the low position before 0.5 sensitivity and rises up. The point where those two curve intercept is called Equal Error Rate(EER) which is 0.0. The result shows the model can successfully detect the dominant and non-dominant hand with 100% accuracy for 4 participants even with small number of training cases.

VI. RESEARCH LIMITATION

Given the performance of the DTW algorithm on the small database, however, a larger database containing more participants should be gathered to better analyze the result. For future studies, more algorithms like the k-nearest neighbor, and deep learning methods like the recurrent neural network, and LSTM should be conducted for the research.

VII. CONCLUSION

In this paper, we proposed a novel traceable dominant hand in-air signature 3D restoration structure for business records and legal documents. The structure restores multiple rounds of the dominant and non-dominant hand signatures into the designated database that can be traced back by server managers once in-air signature-related legal ambiguities happen. The proposed structure can be used to improve the security of current in-air signature applications once constructed. We invited 4 participants to join our research and collected mini-size data set for the purpose of dominant hand recognition using a dynamic time warping algorithm to detect the signal distance between one's dominant and non-dominant hand

signatures. The DTW algorithm has the advantage of being fast and reducing energy consumption compared with other deep learning models, which is especially preferable for portable devices. By using the relative distance from the DTW cost matrix, we can precisely divide the dominant signatures from the non-dominant ones, our algorithm shows 100% accuracy and 0% EER for available participants even with small number of training cases.

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